

PhyST-Net: A Physics-Aware Differentiable Spatio-Temporal Transformer for Constrained Network Forecasting

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Abstract—Spatio-temporal forecasting is a core predictive task for complex graph-structured systems, which is important for operational decision-making in domains such as intelligent traffic routing and resource scheduling. However, existing methodologies frequently blur the temporal distinction between historical observations and future exogenous drivers, while struggling to natively satisfy strict physical operational constraints without relying on naive post-hoc clipping that may weaken gradient signals or create train-inference mismatch. To overcome these limitations, we propose PhyST-Net, a generalized exogenous-aware spatio-temporal graph Transformer framework for domain-constrained network forecasting. Specifically, the framework introduces Adaptive Gaussian-soft-windowed Patching (AGSP) for continuous temporal tokenization and a Relational Spatio-Temporal Manifold Fusion (R-STMF) adapter that utilizes a global token to efficiently decouple macro-trend and local-variation spatial dependencies. Furthermore, it incorporates KAN-enhanced Adaptive Modulation and Conditioning (K-AMC) for non-linear modulation of future exogenous factors, and a Physics-Informed Dynamic Capping Head (PI-DCH) that employs a differentiable barrier gate to promote constraint-aware decoding. The proposed formulation provides a reusable basis for evaluating domain-constrained forecasting systems under diverse operational regimes.

Index Terms—Spatio-temporal forecasting, graph Transformers, exogenous variables, domain-constrained machine learning, network forecasting.

I. INTRODUCTION

SPATIO-TEMPORAL forecasting predicts future states in graph-structured systems (e.g., transportation networks, power grids) where temporal dynamics and spatial interactions evolve jointly. Node-level trajectories are rarely self-driven; rather, they are shaped by historical conditions, future external drivers, and domain-specific feasibility constraints. As illustrated in Fig. 1, complex network signals comprise shared macro-trends driven by exogenous forcing and high-frequency local variations. Consequently, a practical forecasting model must learn continuous temporal representations, infer dynamic graph relations, and integrate future exogenous variables (ExG) without data leakage, all while maintaining operational compatibility.

A. The Forecasting Trilemma in Spatio-Temporal Networks

Despite recent architectural advancements, modern spatio-temporal forecasting frameworks remain hampered by a “Forecasting Trilemma”—an inherent trade-off between Model Capacity, Domain Consistency, and Spatio-Temporal Complexity.

1) *Model Capacity vs. High-Frequency Noise*: Capacity denotes architectural expressive power. While modern Transformers excel at sequence modeling, they are highly sensitive to high-frequency stochastic noise in raw spatio-temporal data. Furthermore, rigid temporal patching exacerbates this by artificially splitting continuous physical regimes. As highlighted by recent trends in complexity-balanced sequence modeling, uniform representational capacity across an entire time horizon is computationally inefficient. The challenge is designing an architecture capable of capturing macro-trends while employing adaptive temporal allocation to filter node-specific noise.

2) *Domain Consistency and Constraint-Aware Operations*: Consistency mandates that a model’s outputs adhere to the fundamental physical laws or operational limits inherent to the target domain. Purely data-driven architectures frequently generate unfeasible predictions—such as negative vehicular speeds or resource usage exceeding an operational upper bound. Such violations severely degrade operator trust in critical infrastructure. Drawing from emerging paradigms in safety-critical generation that utilize output-space guidance, a robust forecasting framework must enforce physical feasibility directly within its decoding pathway, moving beyond the limitations of reactive, post-hoc corrections.

3) *Spatio-Temporal Complexity and Multi-Scale Dynamics*: Complexity involves the management of dense spatial couplings across heterogeneous nodes. In physical networks, nodes interact via multiple simultaneous mechanisms: local topological dependencies (e.g., localized propagation effects, adjacent road segments) and global environmental fronts (e.g., system-wide contextual regimes). Existing models frequently conflate these multi-scale interactions into a singular spatial operator, thereby failing to distinguish localized perturbations from system-wide drivers.

To systematically resolve this trilemma, we propose PhyST-Net, an ExG-aware Spatio-Temporal Graph Transformer designed as a versatile framework for domain-constrained network forecasting. The central contribution of this work is not a single architectural block, but a complete forecasting contract that separates historical representation learning, relational spatio-temporal reasoning, future exogenous conditioning, and constraint-aware decoding into distinguishable pathways. This design directly targets the three tensions above: Adaptive Gaussian-soft-windowed Patching (AGSP) addresses noisy and discontinuous temporal tokenization, Relational Spatio-Temporal Manifold Fusion (R-STMF) separates mixed spatial mechanisms, and the KAN-enhanced Adaptive Modulation and Conditioning (K-AMC) and Physics-Informed Dynamic Capping Head (PI-DCH) connect future drivers with operationally feasible outputs.

TABLE I
TAXONOMY AND COMPARATIVE ANALYSIS OF SPATIO-TEMPORAL FORECASTING MODELS.

Category	Model	Paradigm	Temporal Tokenization	Spatial Interaction	Exogenous Integration	Constraint Awareness
General Transformer Forecasting	Informer [2]	Transformer	Point-wise / ProbSparse	–	Feature concat.	–
	Autoformer [3]	Transformer	Series decomposition	–	Feature concat.	–
	PatchTST [5]	Transformer	Rigid patches	–	–	–
	iTransformer [6]	Transformer	Inverted dimension	Channel mixing	Covariate tokens	–
Spatio-temporal Graph Forecasting	STGCN [9]	ST-GNN	Gated temporal conv.	Static graph conv.	Feature concat.	–
	Graph WaveNet [10]	ST-GNN	Dilated causal conv.	Adaptive graph conv.	Feature concat.	–
	SDG-Net [11]	ST-GNN	Convolutional	Latent dual-graph	–	–
Exogenous-aware Spatio-temporal Forecasting	TimeXer [14]	Hybrid	Rigid patches	Global token / cross-attn.	Variate tokens / cross-attn.	–
	ConFormer [12]	Hybrid	Transformer encoder	Cross-attention	External event emb.	–
	DAG [15]	ST-GNN	Dependency modeling	Directed adaptive graph	ExG-guided dependency	–
	GCGNet [16]	ST-GNN	Generative / recurrent	Consistent alignment	Joint temporal-channel	–
	ExoST [13]	ST-GNN	Convolutional	ExG-driven graph	Select-then-balance	–
Physics-aware Forecasting	GeatFormer [24]	Physics-inf.	Rigid patches	Dual-branch graph	–	Penalty loss
	PGNN-TFT [27]	Physics-inf.	Transformer / TFT	Static domain graph	–	Post-hoc clipping
Ours	PhyST-Net	Hybrid	AGSP	R-STMF	K-AMC	PI-DCH

Notes: “–” indicates that the corresponding mechanism is not explicitly considered. ExG denotes exogenous variables; concat., emb., conv., and attn. denote concatenation, embedding, convolution, and attention, respectively. AGSP, R-STMF, K-AMC, and PI-DCH denote adaptive Gaussian-soft-windowed patching, relational spatio-temporal manifold fusion, KAN-based adaptive modulation conditioning, and physics-informed differentiable constraint head, respectively.

B. Contributions

The primary contributions of this work are summarized as follows:

- **A general ExG-aware graph-forecasting formulation.** We define network forecasting as a multi-modal task where historical states, exogenous variables, and domain constraints are processed through distinguishable pathways, thereby supporting broad architectural generalizability.
- **A continuous temporal tokenization module.** We introduce AGSP, which supersedes rigid patch boundaries with learnable Gaussian soft-windows, preserving event continuity and minimizing boundary artifacts in non-stationary sequences.
- **A modular graph adapter for mixed spatial mechanisms.** We develop R-STMF to disentangle multi-scale spatial dependencies prior to relational propagation, providing a modular contract that accommodates alternative graph predictors.
- **A future-ExG-conditioned decoding path.** We fuse K-AMC and PI-DCH to inject forecast-horizon drivers and structurally promote operational feasibility within the model, bridging latent representation learning with strict operational requirements.

The experimental section later instantiates this formulation on real-world forecasting tasks under diverse operational regimes.

II. RELATED WORK

This section reviews representative research trajectories pertinent to PhyST-Net. Rather than chronicling models chronologically, we categorize prior literature based on the core modeling challenges inherent to ExG-aware spatio-temporal forecasting. A summary of the architectural taxonomies is provided in Table I.

A. Temporal Forecasting and Transformer Architectures

Classical statistical forecasting techniques, including autoregressive and state-space models, retain utility when predicting low-dimensional, approximately stationary processes. However, these foundational assumptions become restrictively fragile in graph-structured systems where node dynamics are highly non-linear and persistently coupled with external forcing. While deep recurrent and convolutional architectures improved non-linear sequence modeling via latent state tracking and local temporal filtering, long-horizon forecasting remains challenging due to error accumulation and vanishing gradients over extended time steps.

Transformer-based forecasters resolve long-range dependency constraints via self-attention, establishing themselves as the dominant paradigm for deep time-series analysis. Early breakthrough models, such as Informer [2], Autoformer [3], and FEDformer [4], focused on reducing the quadratic complexity of attention and decomposing trend-seasonal patterns. Recent variants, utilizing PatchTST-style patching and inverted-dimension attention, significantly reduce computational overhead while enhancing representation learning for multivariate sequences [1], [5], [6]. Nevertheless, temporal patching is predominantly implemented via rigid, fixed-length windows. In highly non-stationary sequences, physical regime transitions and rapid fluctuations rarely align with these artificial boundaries. PhyST-Net advances this trajectory by introducing AGSP, rendering temporal tokenization fully learnable and mathematically continuous.

B. Spatio-Temporal Graph Forecasting

Spatio-temporal graph neural networks (STGNNs) build upon foundational graph convolutions [7] and graph attention mechanisms [8] to capture node interactions. These methodologies are highly effective when cross-node dependencies dictate system behavior, such as in traffic corridors and power grids [9]–[11]. Furthermore, adaptive graph learning relaxes

the restrictive assumption of a static, fully observable topology, allowing models to infer latent relations directly from data and dynamically adjust propagation patterns.

A critical limitation, however, is the widespread reliance on homogeneous propagation protocols across all latent states. In physical systems, spatial dependence is inherently multi-modal; nodes interact through localized proximities, shared environmental influences, and overarching exogenous drivers. A singular adjacency matrix, even if learned dynamically, conflates these distinct mechanisms. R-STMF resolves this by decomposing temporal representations into high-frequency local variations and low-frequency macro-trends, applying targeted, sparse relational propagation to each manifold prior to fusion.

C. Exogenous-Variable-Aware Forecasting

Real-world forecasting is intrinsically dependent on exogenous (ExG) variables. In vertical domains, such as intelligent traffic routing—where models like ConFormer [12] leverage external factors like accidents and regulations—and resource-constrained operational forecasting, ExG features are critical for predictive resilience.

Recent foundational architectures have begun systematically targeting ExG fusion to address inconsistent covariate effects [13]. In time-series modeling, TimeXer [14] empowers Transformers by mapping external factors into variate tokens, enabling cross-attention between endogenous patches and exogenous drivers. Concurrently, DAG [15] introduces dual correlation networks to capture channel-wise dependencies, while in the spatio-temporal graph domain, models like GCGNet [16] employ generative frameworks to align predicted correlations with joint temporal-channel graphs driven by external factors.

While these mechanisms significantly improve accuracy under ideal conditions, they frequently blur the temporal distinction between historical observations and future drivers. Specifically, historical ExG elucidates recent state trajectories, whereas future ExG provides horizon-level modulation. Merging both into a singular feature tensor exacerbates temporal imbalance effects [13] and complicates data availability reasoning at inference time. Consequently, PhyST-Net structurally decouples historical ExG from future ExG. K-AMC leverages the future-ExG pathway to generate latent modulation parameters, enabling forecast-horizon drivers to dynamically condition graph-enhanced states without being erroneously treated as historical priors.

D. Interpretable Conditional Modeling with KANs

Interpretability is paramount in operational forecasting, as dispatchers require insight into the primary drivers shaping a prediction. While attention weights and post-hoc attribution methods offer diagnostic utility, they often fail to reflect the model’s actual computational pathways. Kolmogorov-Arnold Networks (KAN) [19] represent a paradigm shift from traditional perceptrons by modeling transformations via learnable, univariate spline-based edge functions [20]–[23].

By structuring a conditioning network as an additive sum of feature-wise edge functions, the influence of individual

input variables can be analytically inspected through learned basis coefficients and derivatives. PhyST-Net capitalizes on this property within the K-AMC module: future ExG features are mapped to scale and shift parameters via spline functions. While not establishing strict causal identifiability, this architectural alignment provides a transparent, inspectable explanation mechanism far superior to generic dense concatenation.

E. Physics-Informed and Constraint-Aware Forecasting

Physics-informed learning embeds essential domain knowledge into neural architectures via structural priors, penalty losses, or projection operators. In network forecasting, enforcing such constraints is vital, as system targets are strictly bounded by operational capacities. Unfeasible predictions, regardless of aggregate error metrics, can trigger catastrophic downstream decisions.

Contemporary constraint-aware frameworks predominantly rely on physics-guided loss regularization [24], [25] or reactive post-hoc clipping [26], [27]. However, soft penalties do not guarantee compliance, and post-hoc clipping creates a structural disconnect between the trained model and inference behavior. Aligning with the philosophy of structural physics induction [28], the PI-DCH layer internalizes feasibility operations directly within the forecasting head. By utilizing smooth barrier functions to shape predictions toward a dynamic capacity envelope prior to a final non-linear projection, the constraint pathway becomes an explicit, differentiable component of the computational graph.

F. Position of This Work

PhyST-Net synthesizes advancements in Transformer forecasting, adaptive graph learning, and interpretable exogenous conditioning. Its distinguishing factor lies in the rigorous organization of the forecasting interface. Rather than presenting a bespoke, single-domain architecture, this paper formulates a generalized ExG-aware graph prediction framework, subsequently instantiating physically meaningful, domain-specific modules. This methodology positions PhyST-Net as a reusable framework for domain-constrained network forecasting.

III. PROBLEM DEFINITION

We formulate the task as exogenous-variable-aware (ExG-aware) spatio-temporal graph forecasting. This formulation aims to predict future states of a networked system by integrating historical observations, graph-structured dependencies, and external driving factors.

A. Graph Representation and Spatial Dependencies

Let $G = (V, E, A)$ denote a spatio-temporal graph, where $V = \{v_1, \dots, v_N\}$ is the set of N nodes representing physical or logical entities (e.g., wind turbines in a farm or sensors in a city). E is the set of edges representing connections, and $A \in \mathbb{R}^{N \times N}$ is the adjacency matrix encoding spatial relations. In practice, A can be a physical adjacency matrix, a distance-based kernel, or a learned correlation matrix. To maintain flexibility across different domains, we further define M_G

as a set of graph metadata, containing auxiliary information such as node coordinates, static attributes, or environmental descriptors that are not captured by the topology A alone.

B. Temporal Tensors and Exogenous Variables

At a given forecasting origin t , we define the historical target tensor $\mathbf{X}_{t-T+1:t} \in \mathbb{R}^{T \times N \times D_x}$, where T is the look-back window and D_x is the dimension of the target state at each node. Our goal is to predict the future target tensor $\mathbf{Y}_{t+1:t+H} \in \mathbb{R}^{H \times N \times D_y}$ over a forecast horizon H .

A critical challenge in modern forecasting is the integration of exogenous variables (ExG), which we categorize into two distinct types based on their temporal availability:

- **Historical ExG** ($\mathbf{E}_{t-T+1:t}^h \in \mathbb{R}^{T \times D_h}$): These variables represent external conditions (e.g., historical ambient temperature or humidity) that were observed alongside the target state history. They provide contextual information to help the model identify the latent state of the system during the look-back period.
- **Future ExG** ($\mathbf{E}_{t+1:t+H}^f \in \mathbb{R}^{H \times D_f}$): These represent known or forecastable future drivers (e.g., weather forecasts or scheduled operational constraints) that are available for the duration of the forecast horizon. Unlike historical variables, future ExG serves as a conditional driver for the unknown future trajectory.

C. Operational Forecasting Assumptions

To ensure the model is applicable to real-world operational settings, we enforce a strict non-leakage constraint. At inference time t , the model f_θ is only permitted to access information available up to t for target states and historical ExG, and only those future ExG variables that are genuinely known or forecast by external sources (e.g., Numerical Weather Prediction models). This ensures that $\mathbf{E}^f$ does not contain any "future-peeking" information that would be unavailable in a deployed environment.

D. General Optimization Objective

To simplify the notation, we encapsulate all input drivers available at time t into a single input tuple $\mathcal{X}_t$:

$$\mathcal{X}_t \triangleq (\mathbf{X}_{t-T+1:t}, \mathbf{E}_{t-T+1:t}^h, \mathbf{E}_{t+1:t+H}^f, A, M_G). \quad (1)$$

The forecasting task is defined by a parameterized model f_θ which maps $\mathcal{X}_t$ to the predicted horizon $\hat{\mathbf{Y}}_{t+1:t+H}$. For domains where physical laws or operational limits must be satisfied, the model can be decomposed into an unconstrained spatio-temporal backbone $\mathcal{H}_\theta$ and a constraint operator $\mathcal{C}_\Omega$ parameterized by domain knowledge Ω :

$$\hat{\mathbf{Y}}_{t+1:t+H} = \mathcal{C}_\Omega(\mathcal{H}_\theta(\mathcal{X}_t)). \quad (2)$$

The operator $\mathcal{C}_\Omega$ ensures that the final output adheres to feasibility requirements such as non-negativity or capacity limits.

Given a training dataset $\mathcal{D}$, the model parameters θ are optimized by minimizing the Mean Absolute Error (MAE) between the predictions and the ground truth:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}| \cdot H \cdot N} \sum_{(\mathcal{X}, \mathbf{Y}) \in \mathcal{D}} \sum_{h=1}^H \sum_{n=1}^N |f_\theta(\mathcal{X})_{h,n} - \mathbf{Y}_{h,n}|, \quad (3)$$

where the L_1 loss provides robustness against outliers in the spatio-temporal data. This objective provides a general training framework that can be easily extended with additional task-specific regularizers in later stages of the methodology.

IV. METHOD

PhyST-Net is an ExG-aware spatio-temporal graph Transformer designed for network-structured node forecasting with known external drivers and domain constraints. Given the inputs defined in Section III, the model follows the pipeline in Fig. 1. Historical target states and historical ExG are first embedded into node-level temporal representations. AGSP converts the temporal history into continuous tokens. A Transformer backbone learns temporal dependencies. R-STMF serves as a graph adapter that maps temporal node states to graph-enhanced node states. K-AMC injects future ExG into the latent representation. PI-DCH finally decodes the conditioned states into feasible forecasts respecting domain-specific bounds.

A. Overall Framework

For a mini-batch, let $\mathbf{X} \in \mathbb{R}^{B \times T \times N \times D_x}$ denote historical target states, $\mathbf{E}^h \in \mathbb{R}^{B \times T \times D_h}$ denote historical ExG, and $\mathbf{E}^f \in \mathbb{R}^{B \times H \times D_f}$ denote future ExG. The model produces $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times H \times N \times D_y}$. The computation can be written as

$$Z = \Phi_{patch}(\mathbf{X}, \mathbf{E}^h, M_G), \quad (4)$$

$$H_T = \Phi_{temp}(Z), \quad H_G = \Phi_{graph}(H_T, A, M_G), \quad (5)$$

$$H_F = \Phi_{exg}(H_G, \mathbf{E}^f), \quad \hat{\mathbf{Y}} = \Phi_{head}(H_F, \mathbf{E}^f, \Omega), \quad (6)$$

where Φ_{patch} is AGSP, Φ_{temp} is the temporal Transformer encoder, Φ_{graph} is the graph adapter (R-STMF), Φ_{exg} is future-ExG conditioning (K-AMC), and Φ_{head} is the physics-informed prediction head (PI-DCH). This decomposition emphasizes that the architectural backbone is defined on generic graph-structured sequences, while domain-specific knowledge is encapsulated in the input interface and the constraint head.

B. Adaptive Gaussian-soft-windowed Patching

Transformer-based forecasters conventionally mitigate sequence length by aggregating observations into discrete temporal patches. While computationally convenient, fixed non-overlapping patches inevitably induce boundary artifacts when the underlying physical signal transverses a patch boundary. Aligning with emerging principles in complexity-balanced sequence modeling—where representational capacity is dynamically distributed proportional to signal variation—AGSP supersedes rigid patch assignments with learnable Gaussian soft-windows. This differentiable formulation empowers the architecture to autonomously allocate dense receptive fields to

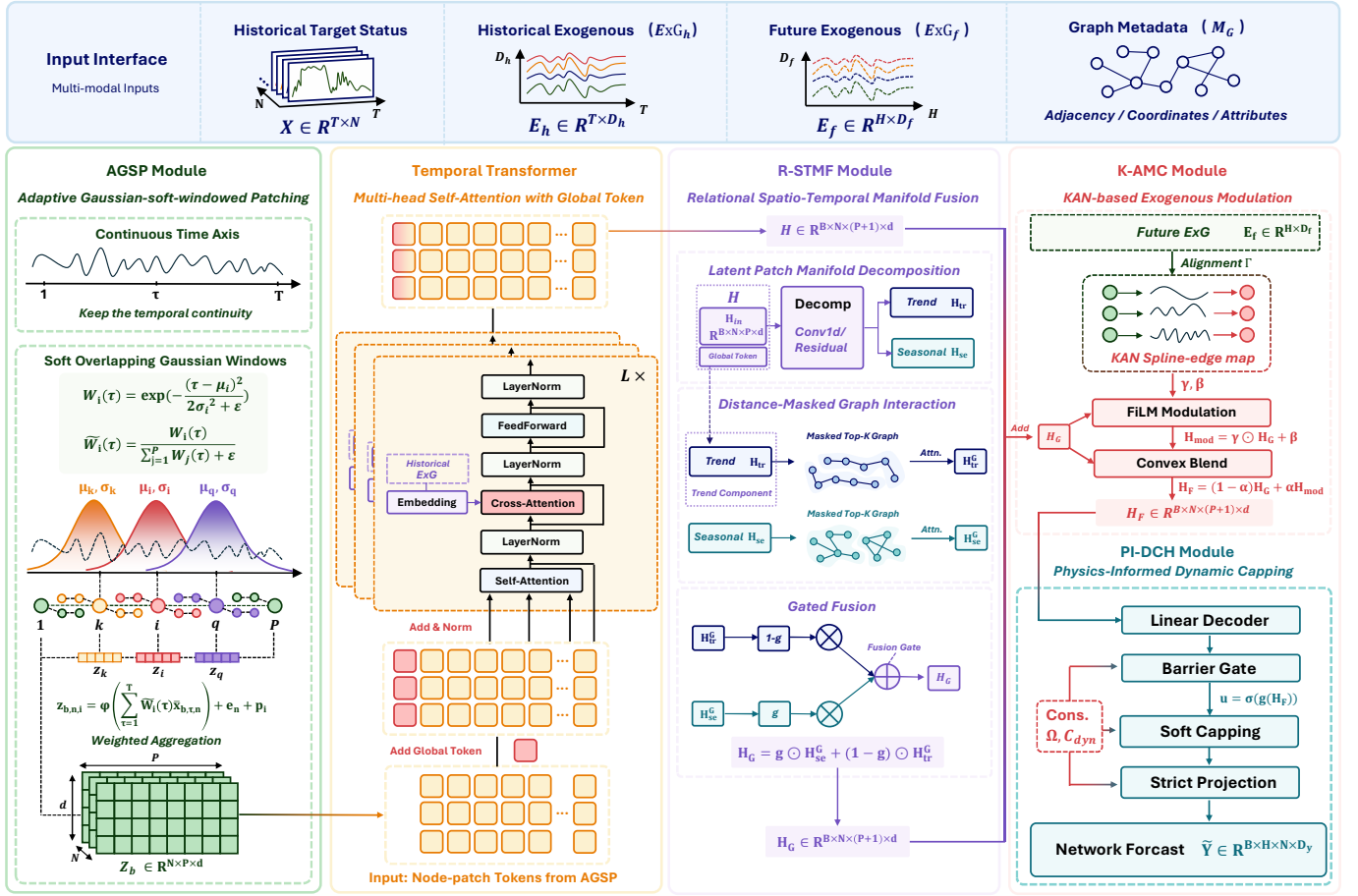


Fig. 1. Overall architecture of PhyST-Net. The framework operates as an exogenous-aware vertical pipeline: (I) Multi-modal input interface; (II) AGSP continuous tokenization; (III) Transformer sequence encoding; (IV) R-STMF relational manifold fusion; (V) K-AMC spline-based conditioning; and (VI) PI-DCH domain-constrained forecasting. All external data (ExG and Metadata) and technical details are organized in peripheral tracks to ensure a clear and interpretable data flow.

highly transient events and broader, smoother fields to stable operational regimes.

For node n and temporal window i , the unnormalized weight at time index $\tau \in \{1, \dots, T\}$ is

$$W_i(\tau) = \exp\left(-\frac{(\tau - \mu_i)^2}{2\sigma_i^2 + \epsilon}\right), \quad (7)$$

where μ_i and σ_i are learnable center and bandwidth parameters. To ensure numerical stability and valid window properties, we parameterize the bandwidth as $\sigma_i = \text{softplus}(\rho_i) + \delta$, where ρ_i is the learnable parameter. The centers μ_i are initialized to uniform grid locations across the historical window to encourage an ordered temporal partition. The weights are normalized across P windows:

$$\tilde{W}_i(\tau) = \frac{W_i(\tau)}{\sum_{j=1}^P W_j(\tau) + \epsilon}. \quad (8)$$

Let $\bar{x}_n(\tau)$ denote the embedded historical state. The token for node n and patch i is

$$z_{n,i} = \psi\left(\sum_{\tau=1}^T \tilde{W}_i(\tau) \bar{x}_n(\tau)\right) + e_n + p_i, \quad (9)$$

where $\psi(\cdot)$ is a linear projection, e_n is a node embedding, and p_i is a temporal patch embedding.

C. Temporal Representation Learning

Adopting the global token philosophy [14] to facilitate efficient long-range context fusion, the token tensor Z is concatenated with a learnable global token $z_{\text{global}} \in \mathbb{R}^{B \times N \times 1 \times d}$ for each node. This global token serves as a sequence-level memory bottleneck, aggregating the macroscopic temporal representation (e.g., shared external regimes or system-wide traffic trends) across all local patches.

We denote the temporal encoding backbone as $\text{STEnc}(\cdot)$. In PhyST-Net, rather than a naive standard Transformer, STEnc incorporates a targeted cross-attention mechanism. Specifically, within each encoder layer, the augmented node-wise token sequence $[Z \parallel z_{\text{global}}]$ first undergoes standard multi-head self-attention. Subsequently, to enrich the macro-trend representation with system-wide context, the global token uniquely cross-attends to the embedded historical exogenous features (e.g., SCADA and time-mark metadata), acting as an information sink:

$$z'_{\text{global}} = z_{\text{global}} + \text{CrossAttention}(z_{\text{global}}, \mathbf{E}^h, \mathbf{E}^h). \quad (10)$$

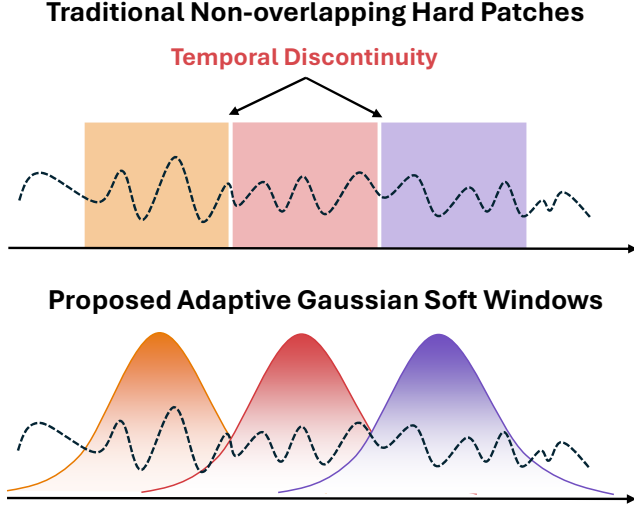


Fig. 2. Mechanics of AGSP. By replacing discrete sequence chunking with continuous, differentiable density windows, the model autonomously mitigates boundary artifacts and preserves temporal continuity. Each soft window is parameterized by a learnable center μ_i and bandwidth σ_i , with overlapping weights dynamically normalized.

This contextually enriched sequence is then processed through the feed-forward networks, yielding the final temporal representation:

$$H_T = \text{STEnc}([Z \parallel z_{\text{global}}]), \quad H_T \in \mathbb{R}^{B \times N \times (P+1) \times d}. \quad (11)$$

Crucially, rather than discarding the global token post-encoding, PhyST-Net structurally leverages it to separate the node's state into two distinct manifolds: a local-variation patch state $U \triangleq H_{T,1:P}$ and a macro-trend global state $R \triangleq H_{T,P+1}$.

D. Relational Spatio-Temporal Manifold Fusion (R-STMF)

R-STMF serves as the graph adapter, specifically designed to capture the multi-modal spatial dependencies inherent in physical networks. Unlike conventional graph operators that treat node features as monolithic vectors, R-STMF recognizes that spatial interactions occur across different temporal scales within the latent space. To address this, we perform a **Latent Patch Manifold Decomposition**.

Let $H_{in} = H_T$ denote the input latent representation. For each node n , we apply a 1D depth-wise convolution along the patch dimension to extract the smoothed trend component H_{tr} , while the residual represents the seasonal (oscillatory) component H_{se} :

$$H_{tr} = \text{GELU}(\text{Conv1d}(H_{in})), \quad (12)$$

$$H_{se} = H_{in} - H_{tr}. \quad (13)$$

This decomposition ensures that the model can independently learn how macro-scale trends and micro-scale seasonal fluctuations propagate through the graph.

1) *Distance-Masked Relational Pruning*: For each manifold $c \in \{se, tr\}$, we compute relational attention scores

across nodes. To enforce physical realism and reduce computational overhead, we introduce a **Distance-Masked Top- K Pruning** strategy. Given a physical distance matrix or prior topology, we define a binary mask $\mathcal{M}_{dist}$, where $\mathcal{M}_{ij} = 1$ only if node j is within a feasible interaction radius of node i . The masked attention score S_c is:

$$S_{c,ij} = \begin{cases} \frac{Q_c(K_c)^\top}{\sqrt{d_m}} & \text{if } \mathcal{M}_{ij} = 1 \\ -\infty & \text{otherwise} \end{cases}, \quad A_c = \text{Softmax}(\text{TopK}(S_c, K_c)). \quad (14)$$

Subsequently, we perform Top- K selection on the remaining scores to ensure a sparse and robust graph routing. Finally, the graph-enhanced seasonal and trend manifolds are recombined using a learnable gated fusion mechanism:

$$g = \sigma(\text{Linear}([H_{se}^G \parallel H_{tr}^G])), \quad (15)$$

$$H_G = g \odot H_{se}^G + (1 - g) \odot H_{tr}^G, \quad (16)$$

where H_c^G denotes the manifold c after spatial propagation via A_c . This dual-branch architecture allows PhyST-Net to model complex spatio-temporal interactions without conflating multi-resolution physical mechanisms.

E. Future-ExG Conditioning (K-AMC)

We refer to this module as K-AMC, short for KAN-enhanced Adaptive Modulation and Conditioning. It serves as a generalized interface for known systemic drivers, injecting future exogenous drivers $\mathbf{E}^f \in \mathbb{R}^{B \times H \times D_f}$ into the latent representations. Aligning with the TimeXer philosophy [14], the macro-trend global token acts as the primary bridge to cross-attend and ingest external factors. We define an alignment operator Γ that maps the future horizon into variate tokens corresponding to the latent states.

The aligned representation $\bar{\mathbf{E}}^f$ is then passed through a Kolmogorov-Arnold Network (KAN) block. Unlike standard MLPs, KANs parameterize transformations using learnable univariate B-spline functions on the edges. For an input feature x , the edge function is defined as:

$$\phi(x) = w_b \text{SiLU}(x) + \sum_{i=1}^k c_i B_i(x), \quad (17)$$

where $B_i(x)$ are B-spline basis functions. The analytical derivative is:

$$\frac{\partial \phi(x)}{\partial x} = w_b \text{SiLU}'(x) + \sum_{i=1}^k c_i B_i'(x). \quad (18)$$

By aggregating these edge functions, the KAN block produces modulation parameters $[\gamma, \beta] = \text{KAN}(\bar{\mathbf{E}}^f)$. The final conditioned latent state is then obtained via element-wise scale and shift, stabilized by a convex blending scheme with hyperparameter $\alpha = 0.1$:

$$H_{mod} = \gamma \odot H_G + \beta, \quad H_F = (1 - \alpha)H_G + \alpha H_{mod}. \quad (19)$$

F. Physics-Informed Dynamic Capping Head (PI-DCH)

PI-DCH instantiates the constraint operator $\mathcal{C}_\Omega$. In mission-critical environments such as smart grids, safety-critical traffic routing, and industrial control systems, deploying predictive

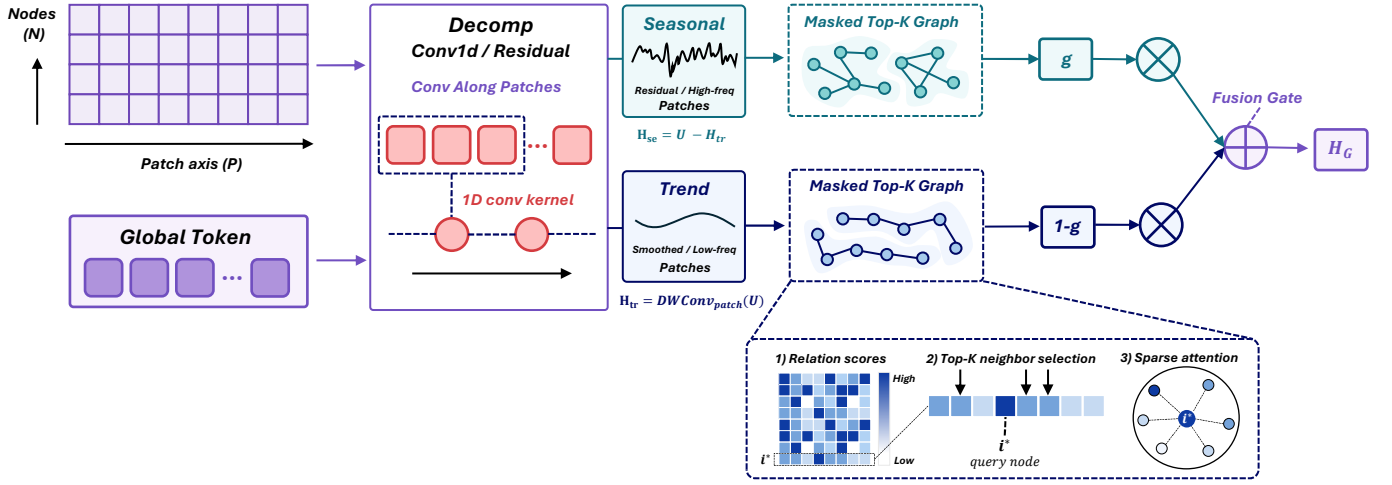


Fig. 3. Detailed architecture of R-STMF. The latent representation is decomposed into trend and seasonal manifolds. Each branch performs distance-masked Top-K graph propagation with branch-specific routing, and the resulting graph-enhanced manifolds are fused through a learnable gate.

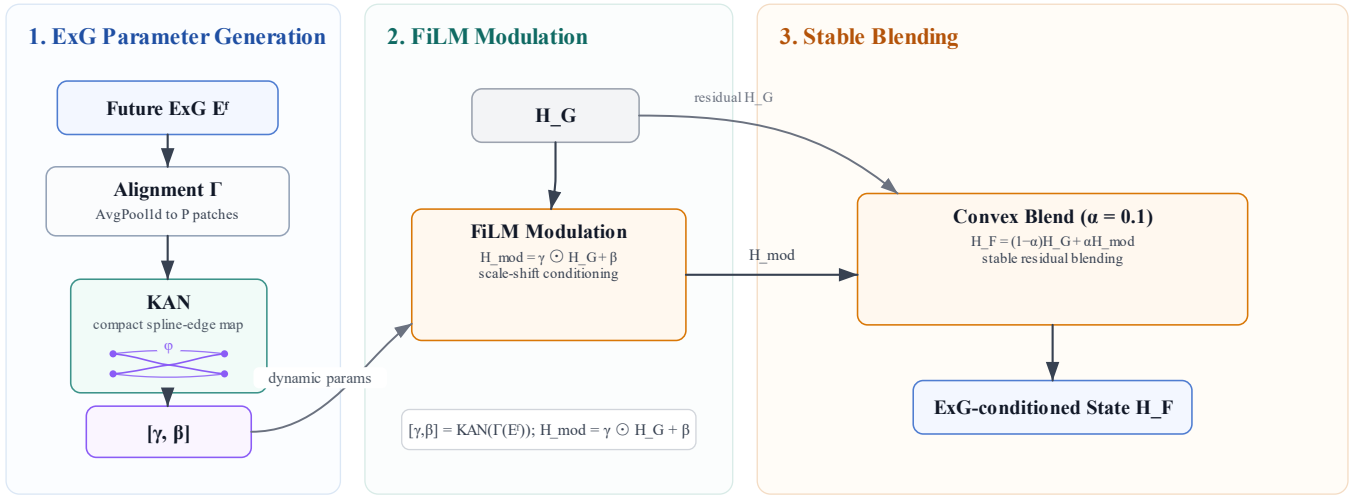


Fig. 4. Mechanics of K-AMC. Future exogenous drivers are aligned to the patch scale and processed by a compact KAN block to generate dynamic modulation parameters. The latent graph state is modulated by FiLM and stabilized through convex blending with the original state.

models mandates rigorous output-space guidance to preclude physically unfeasible trajectories. While hard projection operators (e.g., clipping) guarantee bounded outputs, they may create a mismatch between training and inference, and can severely weaken gradient signals when used naively, hampering the learning process.

To ensure that the model remains physically consistent even under extreme scenarios, we adopt a dual-stage enforcement strategy. While $\hat{Y}_{soft} = \hat{Y} + u \odot (C_{dyn} - \hat{Y})$ provides the differentiable path necessary for optimization via a learned barrier gate $u = \sigma(g(H_F))$, we define the final deployment-ready forecast $\hat{Y}$ through a **Strict Operational Boundary Enforcement** layer:

$$\hat{Y} = \text{Proj}_{\mathcal{K}}(\hat{Y}_{soft}) \triangleq \max(0, \min(\hat{Y}_{soft}, C_{dyn})), \quad (20)$$

where $\text{Proj}_{\mathcal{K}}$ denotes the Euclidean projection onto the dynamic feasible set $\mathcal{K} = [0, C_{dyn}]$. Crucially, this projection is treated as a non-trainable safety layer during inference to guar-

antee 100% compliance with physical limits. By decoupling the differentiable steering ($\hat{Y}_{soft}$) from the strict enforcement ($\hat{Y}$), PhyST-Net achieves the dual goals of stable gradient backpropagation and absolute operational feasibility.

G. Computational Complexity Analysis

A core motivation of PhyST-Net is mitigating the severe computational bottlenecks of dense spatio-temporal Transformers, which conventionally exhibit a time complexity of $\mathcal{O}(N \cdot T^2 + T \cdot N^2)$ for sequence length T and node count N . PhyST-Net structurally resolves this through its decoupled topology.

In the temporal pathway, the AGSP module compresses the raw T -length sequence into P continuous tokens (where $P \ll T$). Consequently, the temporal Transformer encoder's self-attention complexity is reduced from $\mathcal{O}(N \cdot T^2)$ to $\mathcal{O}(N \cdot P^2)$.

In the spatial pathway, standard full-graph attention requires $\mathcal{O}(P \cdot N^2)$ operations. The R-STMF module introduces a

Top- K sparse masking strategy (Eq. 14), restricting message passing to a parameterized neighborhood size K . This bounds the spatial complexity to $\mathcal{O}(P \cdot K \cdot N)$. Thus, the overall self-attention complexity of the PhyST-Net backbone is decoupled and bounded by $\mathcal{O}(N \cdot P^2 + P \cdot K \cdot N)$, rendering the architecture highly scalable for large physical networks and extended forecasting horizons without sacrificing expressive capability.

H. Optimization Objective

To bridge the gap between purely data-driven forecasting and physical feasibility, we minimize a multi-objective loss $\mathcal{L} = \mathcal{L}_{mae} + \lambda_{\Omega} \mathcal{L}_{penalty}$. The penalty term $\mathcal{L}_{penalty}$ structurally regularizes the model to internalize the operational set Ω during backpropagation. Utilizing the positive-part operator $[x]^+ \triangleq \max(0, x)$, we decouple the penalty into two physically meaningful components: a non-negativity constraint and a dynamic capacity limit constraint:

$$\mathcal{L}_{penalty} = \mathcal{L}_{neg} + \mathcal{L}_{cap}, \quad (21)$$

$$\mathcal{L}_{neg} = \frac{1}{|\mathcal{D}|HN} \sum_{\mathcal{D}} \sum_{n,h} [-\hat{Y}_{soft,n,h}]^+, \quad (22)$$

$$\mathcal{L}_{cap} = \frac{1}{|\mathcal{D}|HN} \sum_{\mathcal{D}} \sum_{n,h} [\hat{Y}_{soft,n,h} - C_{dyn,n,h}]^+. \quad (23)$$

This decoupled formulation ensures the model learns to respect physical boundaries natively, thereby reducing the reliance on reactive clipping at inference time. The overall training logic is summarized in Algorithm 1.

V. EXPERIMENTS

This section evaluates PhyST-Net on industrial wind-power forecasting tasks where predictive accuracy and operational feasibility are both essential. To examine its ability to model physical constraints, geographic relations, and weather-driven forcing, we instantiate the framework on distributed wind-power generation networks. Following Section III, each sample contains historical turbine-level targets, historical covariates, forecast-horizon exogenous drivers, and graph metadata. The prediction target is future active power. Unless otherwise stated, all errors are reported on the held-out test split after inverse transformation to physical units.

A. Experimental Protocol

We evaluate two prediction-step settings for each wind-power dataset. These settings are chosen because exogenous forcing, graph propagation, and constraint-aware decoding are all expected to affect the forecast as the prediction length increases. The main comparison covers three wind-power datasets with different spatial scales: SDWPF, HoT, and Kelmarsh.

The dataset selection follows a deliberately constrained evaluation principle. The main benchmark is restricted to wind-power forecasting because wind generation simultaneously exposes the three modeling requirements targeted by PhyST-Net: turbine-level physical bounds, geographically structured

Algorithm 1 PhyST-Net Forward Pass

Require: Input tuple $\mathcal{X}_t = (\mathbf{X}, \mathbf{E}^h, \mathbf{E}^f, A, M_G)$, Domain knowledge Ω .

Ensure: Feasible network forecast $\hat{\mathbf{Y}}$.

- 1: *{Phase I: AGSP Continuous Tokenization}*
 - 2: Initialize learnable Gaussian parameters μ_i, σ_i for P tokens using metadata M_G .
 - 3: **for** each token $i \in \{1, \dots, P\}$ **do**
 - 4: Compute soft-window weights $W_i(\tau) = \exp\left(-\frac{(\tau - \mu_i)^2}{2\sigma_i^2 + \epsilon}\right)$.
 - 5: Aggregate $\mathbf{X}$ and $\mathbf{E}^h$ via $W_i(\tau)$ to form continuous token $Z_{:,i,:}$.
 - 6: **end for**
 - 7: *{Phase II: Temporal Encoding ($\mathcal{H}_{\theta}$ part 1)}*
 - 8: Process Z via Multi-Head Self-Attention to capture temporal dependencies $\rightarrow H_T$.
 - 9: *{Phase III: R-STMF Graph Adaptation ($\mathcal{H}_{\theta}$ part 2)}*
 - 10: Decompose $H_{in} = H_T$ into seasonal H_{se} and trend H_{tr} manifolds.
 - 11: Apply Top- K sparse masking to A based on spatial similarities and physical distances.
 - 12: Propagate H_{se} and H_{tr} over the masked graph and fuse $\rightarrow H_G$.
 - 13: *{Phase IV: K-AMC Exogenous Modulation ($\mathcal{H}_{\theta}$ part 3)}*
 - 14: Align horizon-wise $\mathbf{E}^f$ to patch-wise representation $\bar{\mathbf{E}}^f$.
 - 15: Compute B-spline edge functions via KAN block to generate $[\gamma, \beta]$.
 - 16: Modulate latent states: $H_F \leftarrow (1 - \alpha)H_G + \alpha(\gamma \odot H_G + \beta)$.
 - 17: *{Phase V: PI-DCH Constraint Decoding ($\mathcal{C}_{\Omega}$)}*
 - 18: Decode unconstrained prediction $\hat{\mathbf{Y}} = \text{Linear}(H_F)$.
 - 19: Shape prediction softly: $\hat{\mathbf{Y}}_{soft} \leftarrow \hat{\mathbf{Y}} + u \odot (C_{dyn} - \hat{\mathbf{Y}})$.
 - 20: Project to feasible set: $\hat{\mathbf{Y}} \leftarrow \text{Proj}_{\mathcal{K}}(\hat{\mathbf{Y}}_{soft})$.
 - 21: **return** $\hat{\mathbf{Y}}$.
-

propagation, and weather-driven future forcing. Within this domain, SDWPF provides a large industrial network for scalability and spatial interaction, HoT offers a medium-size farm with long chronological coverage and moderate missingness, and Kelmarsh supplies a smaller but exogenous-rich setting for testing whether the model can benefit from richer meteorological descriptors. This combination avoids mixing incomparable target semantics in the primary leaderboard while still covering different graph sizes, covariate dimensions, sampling intervals, and prediction-step lengths.

SDWPF is a Chinese wind-farm dataset from Shandong with 134 turbines sampled every 15 minutes over more than two years. The loader supports both the historical HDF5 files and the normalized CSV layout, including turbine-level power targets, SCADA measurements, numerical weather prediction (NWP) variables, and turbine positions. The current SDWPF configuration uses average wind speed and average wind direction as meteorological drivers, together with standardized turbine coordinates for spatial encoding. HoT is the Hill of Towie wind-farm dataset from Scotland, with 21 turbines, 192794 timestamps, five exogenous variables, and 10-minute records from 2020-01-01 to 2024-09-01; its target missing

TABLE II
EXPERIMENTAL DESIGN FOR REPRESENTATIVE WIND-POWER TASKS.

Dataset	N	D_f	Freq.	T	H
HoT	21	5	10 min	576	432
HoT	21	5	10 min	1920	1440
Kelmarsh	6	12	10 min	576	432
Kelmarsh	6	12	10 min	1920	1440
SDWPF	134	2	15 min	384	288
SDWPF	134	2	15 min	1280	960

rate is 1.83%. Kelmarsh is a UK wind-farm dataset with six turbines, 157536 timestamps, 12 exogenous variables, and 10-minute records from 2016-01-03 to 2018-12-31; the processed target matrix contains no missing values. All three datasets follow a common wind-task interface with target, SCADA, NWP, and position files.

All datasets are split chronologically to avoid temporal leakage. The first 70% of samples are used for training, the next 10% for validation and early stopping, and the final 20% for testing. Missing target values are filled before scaling by forward filling followed by zero filling for any remaining entries; when available, the original missingness can still be retained as a mask. Standardization parameters for power targets are fitted on the training segment and reused for validation and testing. Meteorological variables are scaled consistently across historical SCADA and forecast-horizon NWP inputs, and turbine coordinates are standardized before entering the spatial module. Calendar encodings include month, day, weekday, hour, and the native intra-hour bucket for each sampling interval.

The repository also contains loaders and exploratory summaries for HK Rooftop PV and ASHRAE building load. These datasets are not included in the main leaderboard because their target semantics and operational envelopes differ from turbine-level active power. Instead, they are used in the ablation study to test whether the proposed modules remain useful when the same forecasting contract is instantiated on related constrained-energy tasks.

Table II summarizes the representative configurations used in the main comparison. HoT and Kelmarsh are sampled every 10 minutes, while SDWPF is sampled every 15 minutes. For 10-minute datasets, the two prediction lengths are $H = 432$ and $H = 1440$, corresponding to 3 and 10 days ahead, respectively. For SDWPF, the step counts are adjusted to the 15-minute sampling interval, yielding $H = 288$ and $H = 960$ for the same physical durations. This design reports the task by prediction steps while keeping the physical duration comparable across datasets.

PhyST-Net uses a 3-layer encoder, one decoder layer, hidden size 256, feed-forward size 512, eight attention heads, dropout 0.1, and Adam optimization. Training is capped at 50 epochs with early stopping patience 5 and gradient clipping at 1.0. The loss combines mean-squared prediction error with a high-output penalty weighted by 2.0 to emphasize rated-power regimes. In the full model, AGSP, R-STMF, and K-AMC are enabled throughout the main comparison. The ablation scripts keep the same data split, optimizer, prediction length, and

model width, changing only the target module switch.

We compare PhyST-Net with strong recent time-series and spatio-temporal baselines, including TimeXer, CrossFormer, PatchTST, iTransformer, Amplifier, DAG, DUET, and TimeKAN. For models that do not natively consume exogenous wind variables, we use a lightweight convolutional exogenous adapter so that all methods are evaluated with the same historical-power and future-NWP information. The primary metrics are mean absolute error (MAE) and root mean square error (RMSE), where lower values indicate better predictive accuracy.

B. Main Results

Table III summarizes the representative benchmark results. PhyST-Net achieves the best MAE in all six settings and the best RMSE in four of them. The gains are most pronounced on Kelmarsh, where PhyST-Net reduces MAE by 30.99% at $H = 1440$ and 25.05% at $H = 432$ relative to the strongest competing baseline. On SDWPF, it also maintains clear MAE reductions of 21.44% at $H = 960$ and 14.06% at $H = 288$. On HoT, PhyST-Net gives the lowest MAE for both prediction lengths, while DAG and TimeKAN remain competitive in RMSE. Overall, the results indicate that PhyST-Net is most effective when forecasting requires non-local spatial interaction and future exogenous information rather than near-term autoregressive continuity alone.

Fig. 5 is reserved for the final result visualization. The intended figure emphasizes relative improvement rather than only absolute errors, making the comparison less sensitive to dataset-specific power scales. It should also expose the two settings where RMSE competition is closest on HoT, because these cases clarify that PhyST-Net’s main advantage lies in consistent MAE reduction and robust long-horizon behavior rather than uniformly dominating every error norm.

C. Ablation Study Across Multiple Domains

To assess the contribution of each module across forecasting contexts, we conduct ablation studies on ASHRAE building energy, HK Rooftop PV, and SDWPF wind power. These datasets stress different aspects of the model: ASHRAE reflects building-level load dynamics with strong diurnal structure, HK Rooftop PV captures weather-driven solar variability, and SDWPF represents industrial wind power with rich inter-turbine dependencies. Together, they test adaptive patching, relational manifold modeling, and exogenous conditioning under diverse operating regimes.

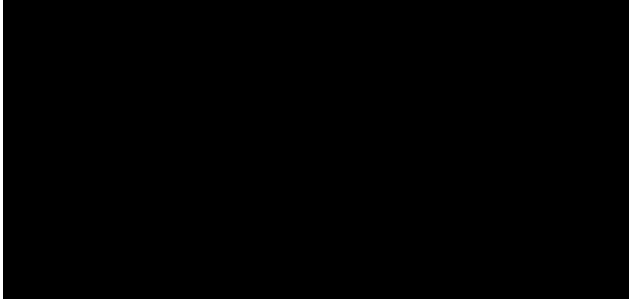
Tables IV–VI report the full model and three controlled variants. The anchor is the complete PhyST-Net configuration. The w/o AGSP variant removes adaptive Gaussian-soft-windowed patching, w/o K-AMC removes KAN-enhanced adaptive meteorological conditioning, and w/o R-STMF removes relational spatio-temporal manifold fusion. We report MAE, RMSE, WAPE, and R^2 for each dataset.

Across the three datasets, the anchor model achieves the best or near-best performance on most metrics. The ablation patterns are domain-dependent: removing R-STMF causes the largest degradation on ASHRAE, removing K-AMC is most

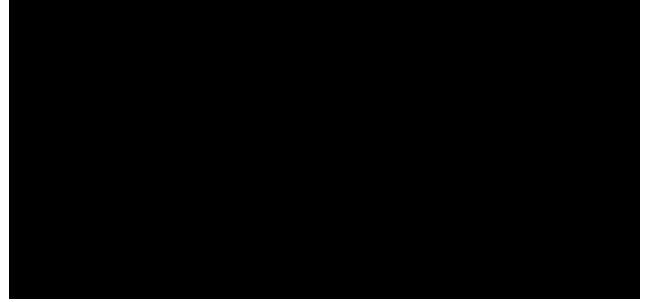
TABLE III
MAIN FORECASTING RESULTS ON REPRESENTATIVE WIND-POWER SETTINGS.

Dataset	H	PhyST-Net		TimeXer		CrossFormer		PatchTST		iTransformer		Amplifier		DAG		DUET		TimeKAN	
		MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
HoT	1440	267.32	468.23	377.48	556.27	<u>323.40</u>	480.46	372.19	529.96	428.58	647.79	394.77	537.81	374.66	530.49	359.42	508.00	325.33	<u>477.37</u>
HoT	432	258.55	<u>449.75</u>	370.98	536.43	363.96	526.93	369.48	504.82	312.79	496.73	396.91	549.22	<u>276.90</u>	445.89	363.62	509.49	343.95	501.63
Kelmarsh	1440	144.55	208.41	231.16	328.12	280.51	410.72	349.92	477.61	<u>209.45</u>	<u>283.89</u>	382.84	527.38	306.12	410.92	264.88	361.46	337.94	445.28
Kelmarsh	432	138.67	200.07	202.08	285.36	277.38	397.51	299.90	404.94	186.61	<u>273.30</u>	348.30	475.12	<u>185.01</u>	276.33	290.11	377.49	329.03	433.20
SDWPF	960	129.85	215.72	198.01	281.86	185.61	256.27	249.13	325.73	<u>165.29</u>	<u>233.61</u>	246.56	350.77	167.78	246.62	195.13	277.77	OOM	OOM
SDWPF	288	126.30	200.86	164.12	244.59	223.18	300.65	223.91	321.39	165.21	237.90	266.87	376.54	<u>146.96</u>	<u>230.52</u>	217.38	293.28	216.25	288.11

Notes: Bold and underlined values indicate the best and second-best results, respectively. OOM denotes out-of-memory failure under the same evaluation environment. Full prediction-step results are kept as supplementary evidence.



(a) MAE reduction over the strongest baseline



(b) RMSE comparison across prediction steps

Fig. 5. Planned visualization for the main benchmark. Panel (a) should report the relative MAE reduction of PhyST-Net against the best non-PhyST-Net baseline for each dataset- H pair. Panel (b) should plot RMSE across prediction steps to show where PhyST-Net, DAG, TimeKAN, and other competitive baselines differ most clearly.

TABLE IV
ABLATION STUDY ON ASHRAE BUILDING ENERGY (METER0 SITE0).

Variant	MAE	RMSE	WAPE	R^2
PhyST-Net (anchor)	50.151	91.591	0.1393	0.9512
w/o AGSP	52.905	92.839	0.1470	0.9498
w/o K-AMC	55.281	97.646	0.1536	0.9445
w/o R-STMF	58.286	102.650	0.1619	0.9387

Notes: Removing R-STMF produces the largest degradation (MAE +16.2%, RMSE +12.1%).

TABLE V
ABLATION STUDY ON HK ROOFTOP PV FORECASTING.

Variant	MAE	RMSE	WAPE	R^2
PhyST-Net (anchor)	2201.51	6742.61	0.4064	0.7494
w/o AGSP	2318.02	7209.00	0.4279	0.7135
w/o K-AMC	2258.50	6841.35	0.4169	0.7420
w/o R-STMF	2351.25	7212.29	0.4340	0.7133

Notes: Removing R-STMF yields the largest MAE increase (+6.8%).

TABLE VI
ABLATION STUDY ON SDWPF FORECASTING ($H = 288$).

Variant	MAE	RMSE	WAPE	R^2
PhyST-Net (anchor)	123.14	201.83	0.3730	0.7635
w/o R-STMF	124.19	202.90	0.3762	0.7610
w/o K-AMC	126.09	206.53	0.3820	0.7524
w/o AGSP	123.85	204.37	0.3691	0.7576

Notes: Removing K-AMC produces the largest MAE and RMSE degradation (+2.4% and +2.3%, respectively).

harmful on SDWPF, and removing AGSP substantially affects HK Rooftop PV. These results indicate that all three modules contribute to the forecast, while their relative importance depends on the physical dynamics of each domain.

Fig. 6 complements the tabular ablations by making the domain dependence of each module explicit. The expected pattern is not a single universally dominant component: R-STMF should be most visible when cross-node relational structure is strong, K-AMC should matter when forecast-horizon weather drivers control the target trajectory, and AGSP should help when sharp temporal transitions or boundary artifacts dominate the signal.

D. Discussion of the Evaluation Scope

The benchmark results show that PhyST-Net is strongest in the evaluated multi-step prediction regimes, where future meteorological drivers and spatial propagation have sufficient time to influence the target trajectory. We therefore emphasize the operating regime most aligned with the design of PhyST-Net: constrained graph forecasting under exogenous forcing.

The ablation evidence spans multiple domains and shows that adaptive patching, relational manifold modeling, and exogenous modulation improve forecasting across building energy, rooftop PV, and wind-power tasks. These results complement the cross-dataset wind-power comparison on SDWPF, HoT, and Kelmarsh.

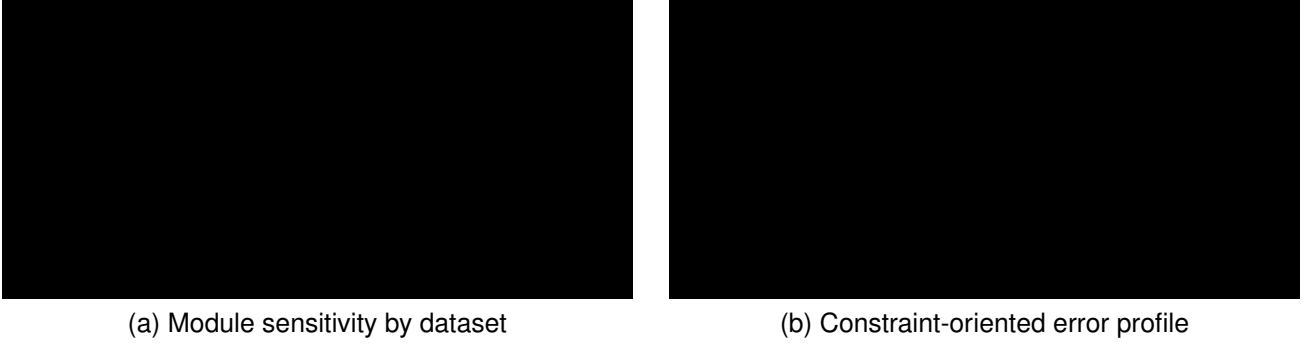


Fig. 6. Planned visualization for module-level analysis. Panel (a) should summarize the relative degradation caused by removing AGSP, R-STMF, or K-AMC on ASHRAE, HK Rooftop PV, and SDWPF. Panel (b) should visualize constraint-sensitive behavior, such as near-capacity errors and lower-bound violations, to connect quantitative ablation results with the PI-DCH design.

VI. DISCUSSION

This section discusses the implications of our results, focuses on the scalability of the proposed framework, and analyzes its generalization capabilities across diverse industrial network environments.

A. Domain Generalization across Industrial Networks

The experimental results demonstrate that PhyST-Net is robust across the evaluated domains, covering wind generation in the main comparison (SDWPF, HoT, Kelmars) and auxiliary ablations on HK Rooftop PV and ASHRAE building energy. This domain-independent behavior stems from the modular separation between spatial routing, non-linear exogenous modulation, and physical decoding. Specifically, in generation-side forecasting (wind and rooftop PV), the physical constraints primarily restrict output within $[0, P_n]$, preventing unphysical negative generation and mechanical overload predictions. In demand-side building-energy forecasting, the constraint head enforces non-negativity and dynamically tracks historical peaking envelopes. K-AMC effectively models weather-driven factors (e.g., wind speed, solar irradiance, temperature) by using B-spline KAN parameterization, bypassing standard multi-layer perceptron (MLP) limitations in localizing high-frequency exogenous shifts.

B. Computational Complexity and Scalability

A key barrier in deploying spatio-temporal graph Transformers on large-scale industrial networks is the computational cost, which conventionally scales quadratically with both the number of network nodes N and the sequence length T . PhyST-Net addresses these limits through two design features:

- **Linear Spatial Routing:** R-STMF utilizes a global token as an information sink. Instead of computing direct node-to-node attention with $O(N^2)$ complexity, R-STMF employs cross-attention between local node representations and the global token. This reduces spatial interaction complexity to $O(N)$ per layer, enabling scalability to large networks like SDWPF ($N = 134$).
- **Continuous Tokenization:** AGSP aggregates raw temporal steps into continuous patches. This tokenization

reduces the effective temporal sequence length from T to T/P , where P is the window stride, leading to significant savings in the downstream temporal Transformer layers.

C. Gradient Flow under Domain Constraints

Traditional methods enforce operational envelopes via post-hoc hard clipping, i.e., $\hat{Y} = \max(0, \min(P_n, Y_{raw}))$. While simple, hard clipping zeroes out gradients for out-of-bounds predictions, causing gradient vanishing during backpropagation and hindering the learning of extreme events. In contrast, the Physics-Informed Dynamic Capping Head (PI-DCH) implements a differentiable barrier gate. By employing a double-sigmoid soft truncation, PI-DCH keeps gradient paths active even when predictions approach or exceed physical limits. This ensures that the model receives continuous feedback during training, optimizing predictive accuracy in near-capacity operational regimes.

D. Limitations and Future Work

While PhyST-Net provides a unified framework for constrained forecasting, certain limitations remain. First, the graph structure in R-STMF is currently static. In real-world industrial networks, topology can change dynamically due to transmission line maintenance or sensor failures. Extending R-STMF to dynamic graphs represents an important research direction. Second, the current formulation is deterministic. Providing calibrated probabilistic interval forecasts under physical constraints would further improve operational decision-making.

VII. CASE STUDIES

This section provides qualitative analysis through four distinct case studies to illustrate how the individual components of PhyST-Net address specific operational challenges in industrial network forecasting.

A. Case A: Differentiable Upper Capping under Extreme Meteorological Events

In industrial generation networks, such as wind farms or solar PV fields, extreme weather events (e.g., storms or peak solar noon) can drive predictions beyond the physical capacity

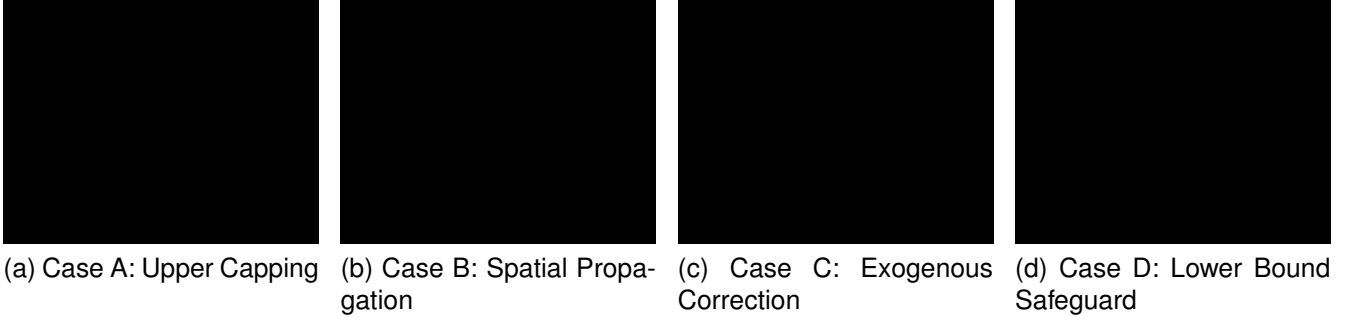


Fig. 7. Visualization of qualitative case studies demonstrating the effectiveness of PhyST-Net modules in diverse operational scenarios. (a) PI-DCH prevents over-capacity predictions (> 1.0) during extreme weather events. (b) R-STMF tracks spatial attention delay as meteorological fronts propagate across nodes. (c) K-AMC B-splines learn non-linear local environmental corrections. (d) PI-DCH enforces a strict non-negativity constraint in low-speed/night regimes.

of the system. As illustrated in Fig. 7a, under high wind-speed conditions on the HoT dataset, standard Transformer backbones produce active power predictions that exceed the rated capacity (normalized power > 1.0). If a post-hoc hard clip is applied to enforce the upper limit, the gradient becomes zero, which restricts learning. In contrast, the PI-DCH module employs a differentiable barrier gate that smoothly caps predictions at 1.0. This soft capping mechanism preserves the gradient flow throughout training, enabling the network to learn optimal weights even in near-saturation regimes.

B. Case B: Spatio-Temporal Propagation and Advection Delay

Physical phenomena propagating across industrial networks (e.g., wind front advection in wind farms or cloud movement over distributed solar networks) exhibit clear spatio-temporal phase lags. Fig. 7b visualizes the spatial attention weights computed by the R-STMF adapter across the SDWPF turbine network. During a meteorological front event, R-STMF dynamically shifts attention from upstream nodes to downstream nodes in accordance with the wind speed and direction. By utilizing the global token as an intermediate information sink, the model captures this network-wide propagation lag without incurring the quadratic computational cost of full pairwise spatial attention.

C. Case C: Non-linear Exogenous Correction for Topographical Effects

Industrial networks are frequently deployed in complex terrains where local micro-climates or topographical shadows (e.g., turbine wake shadows or solar panel tilt occlusions) lead to non-linear deviations from regional forecasts. Fig. 7c depicts the conditioning scaling factor learned by the K-AMC module relative to numerical weather prediction (NWP) inputs on the Kelmarsh dataset. Unlike standard linear or MLP-based modulation schemes that perform uniform scaling, K-AMC utilizes B-spline parameterization to learn highly localized, piece-wise non-linear correction curves. This allows the model to adjust predictions for specific wind speed ranges where terrain-induced turbulence is most prominent.

D. Case D: Zero-Generation Safeguard during Low-Operational States

At the lower bound of operations, industrial generators do not produce power (e.g., wind turbines below the cut-in wind speed, or solar panels at night). autocovariance fluctuations in standard mean-squared error (MSE) backbones often cause models to predict unphysical negative power values. As shown in Fig. 7d, during low wind-speed intervals on the SDWPF dataset, the baseline model outputs negative generation values (down to -2%). While small, these negative predictions accumulate across large networks and violate physical laws. PI-DCH resolves this by enforcing a differentiable lower-bound barrier. This caps the prediction at exactly 0.0 while ensuring that the backpropagated gradient remains informative, preventing negative power leakage without destabilizing training.

VIII. CONCLUSION

This paper presented PhyST-Net, a physics-aware differentiable spatio-temporal Transformer for constrained network forecasting with historical observations, future exogenous drivers, graph metadata, and operational bounds. Rather than treating these inputs as a single fused tensor, PhyST-Net defines a structured forecasting contract: AGSP provides continuous temporal tokenization, R-STMF separates relational spatio-temporal mechanisms, K-AMC injects forecast-horizon drivers through non-linear adaptive modulation, and PI-DCH promotes feasible decoding through a differentiable constraint head.

The experimental design focuses the main leaderboard on wind-power forecasting, where physical capacity limits, geographic propagation, and meteorological forcing are all present. Across SDWPF, HoT, and Kelmarsh, PhyST-Net achieves the best MAE in all representative prediction-step settings and the best RMSE in most of them, with especially clear gains on Kelmarsh and SDWPF. Auxiliary ablations on ASHRAE building energy, HK Rooftop PV, and SDWPF further show that the proposed modules contribute differently across domains: relational manifold modeling, adaptive temporal patching, and future-driver conditioning become most important under different physical regimes.

Several directions remain open. The current graph interface is primarily static, while deployed infrastructure may expe-

rience topology changes, sensor outages, and maintenance-induced distribution shifts. The present formulation is also deterministic; extending PI-DCH to calibrated probabilistic forecasts under hard operational envelopes would better support risk-aware scheduling. Finally, larger-scale deployment should evaluate how the forecasting contract behaves under richer NWP products, additional energy assets, and online adaptation scenarios.

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